Chapter 1: Overview of Databases

Data models

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2. Network data model
3. Entity relationship model

1. Introduction

Data model is an abstraction of real environment, representing data at conceptual level.

Some popular data models:

- Network data model
- Hierarchical data model
- Entity-relationship data model
- Relational data model
- Object-oriented data model

2. Network data model (1)

- *Network Data Model*, also known as network model, is a model represented by a directed graph.
- Building blocks:
  - Record
  - Record type
  - Relationship (Set type)
  - Cardinality

2. Network data model (2)

- **Record** : describes an object in the real world.

('NV001','Nguyen Lam','Nam','October 10, 1970','Dong Nai')

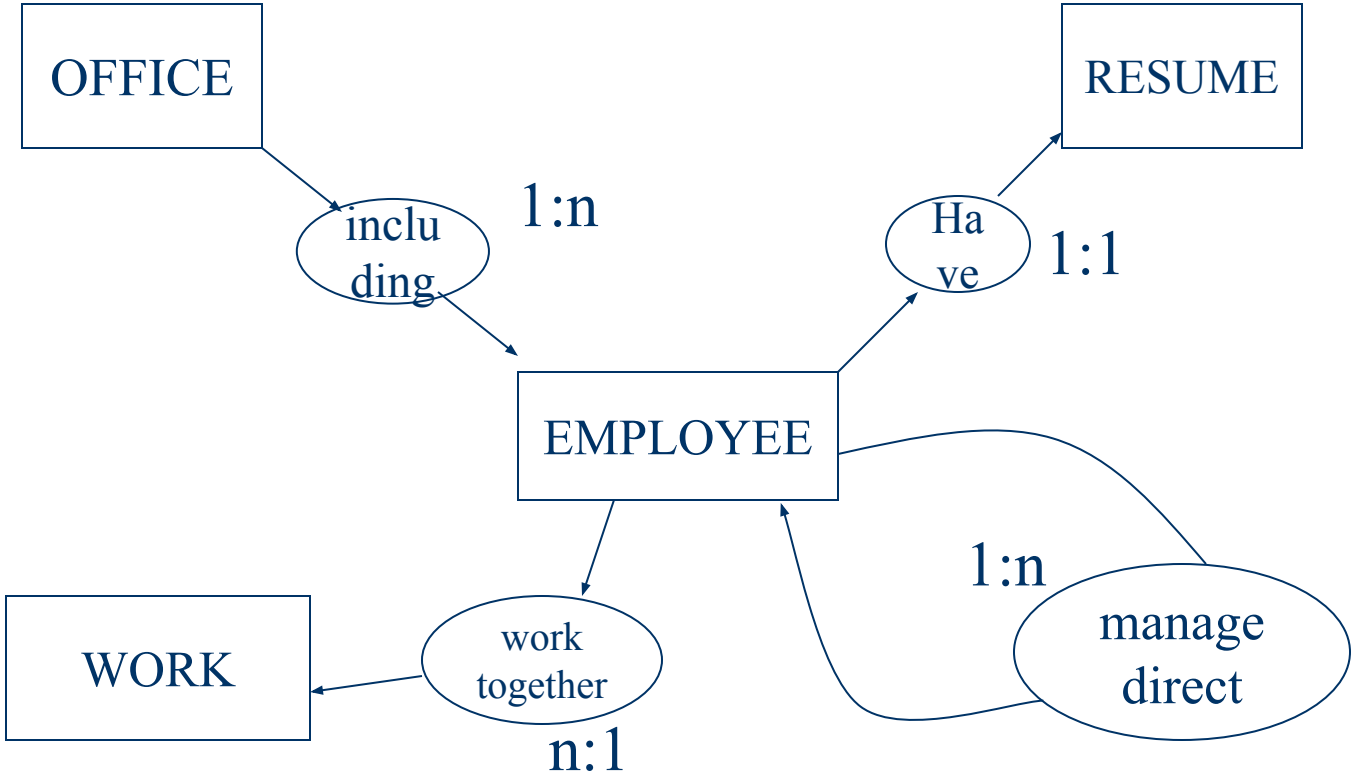
- **Record type** : is a set of records with the same properties. For example: NHANVIEN, EMPLOYEE
  - Symbol:

- **Relationship (Set type)** : describes the association between a master record type and a member record type.
  - Symbol:

2. Network data model (3)

- **Cardinality:** indicates the number of records participating in the relationship
  - **(1:1)** (one-to-one): each record of the master record type is associated with exactly one record of the member record type.
  - **(1:n)** (one-to-many): each record of the master record type is associated with one or more member records.
  - **(n:1)** (many-to-one): many records of the master record type are associated with exactly one record of the member record type.
  - (Recursive): a master record type can also be a member record type with itself. This type of relationship is Recursive

2. Network data model (4)



2. Network data model (5)

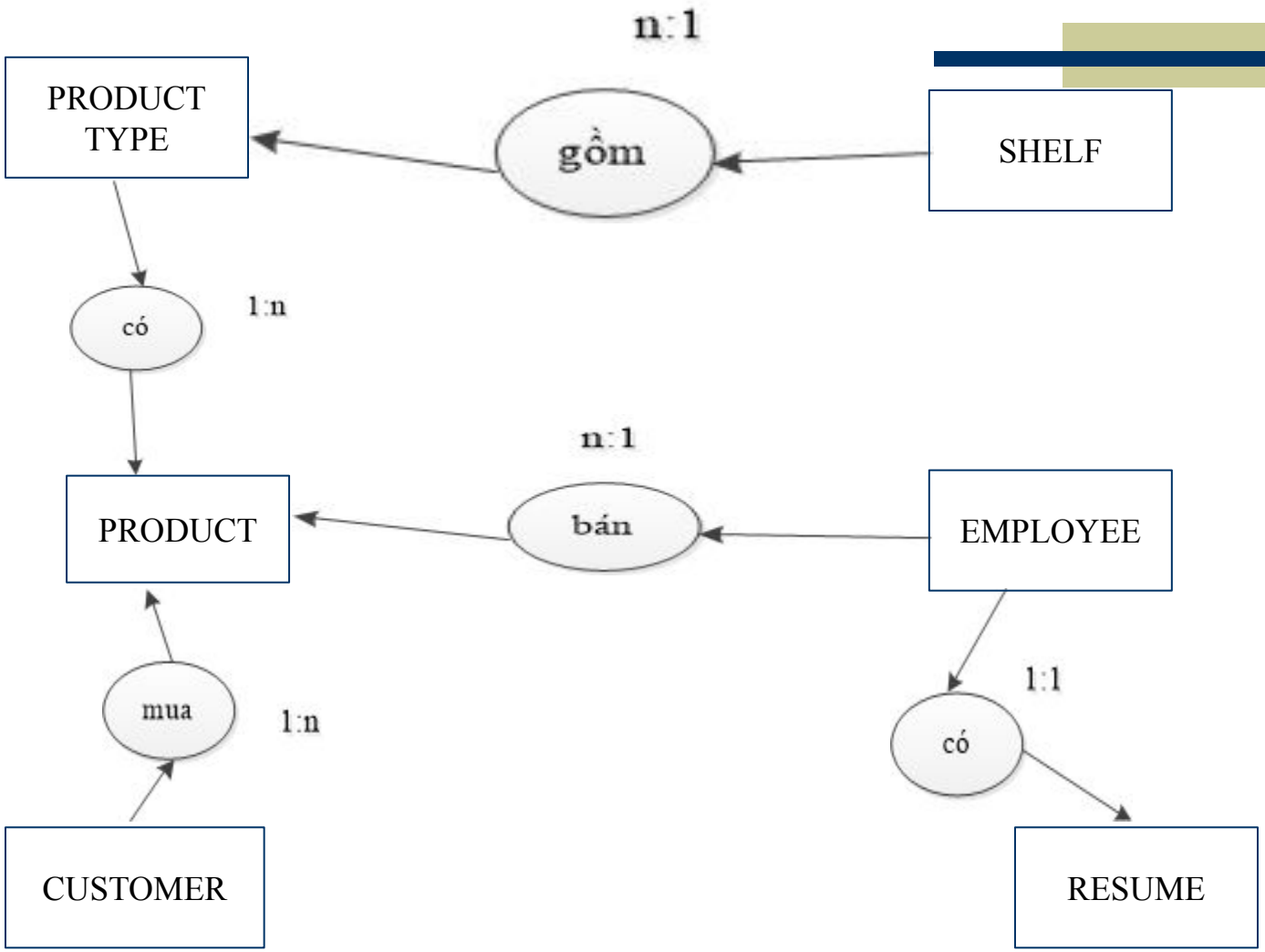
- Network data model:
  - Relatively simple
  - Easy to use
  - Not suitable for representing large-scale databases
  - Poor semantic expression ability

2. Network data model (6)

- **Exercise** :

Building a network data model for a sales management database in a supermarket

(Teacher describes the business process, identifies requirements)


n:1
PRODUCT TYPE
gồm
SHELF
có
1:n
n:1
PRODUCT
bán
EMPLOYEE
mua
1:n
1:1
có
CUSTOMER
RESUME

3. Entity relationship model

3.1 Introduction
3.2 Entity type, entity
3.3 Attributes of entity type
3.4 Keys of entity type
3.5 Relationship type, relationship
3.6 Attributes of relationship type
3.7 Cardinality
3.8 Extended ER model

3.1 Introduction

- *Entity-Relationship Model* (ER) was introduced by CHEN in 1976.
- ER model is widely used in conceptual level data design.

3.2 Entity type

- **Definition** : Entity Type is a specific type of real-world object or thing that needs to be managed.
- **For example** : HOCVIEN, LOP, MONHOC, …
- **Symbol** :

HOCVIEN

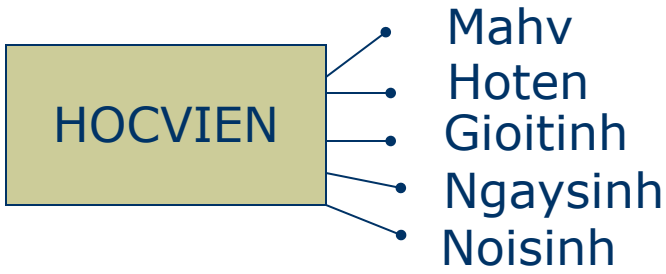
LOP

3.2 Entity

- **Definition** : An entity is an instance of an entity type.
- **For example** : Entity type HOCVIEN has the following entities:
  - ('HV001', 'Nguyen Minh', '1/2/1987','Male')
  - ('HV002', 'Tran Nam', 'February 13, 1987', 'Nam')

3.3 Attributes of entity type

- **Definition** : attributes are characteristics of an entity type.
- **For example** : The entity type HOCVIEN has the following attributes: Student ID, full name, gender, date of birth, place of birth.
- **Symbol** :



3.3 Attribute types (1)

- ***Simple-valued***: each entity has only one value for each attribute.
  For example: Mahv, Hoten
- ***Composite***: an attribute can be made up of multiple components.
  For example: DCHI(SONHA, DUONG, PHUONG, QUAN) or HOTEN(HO, TENLOT, TEN).
- ***Multi-valued***: an attribute can have multiple values for an entity.
  For example: BANGCAP is denoted by {BANGCAP}

3.3 Attribute types (2)

- Note: composite and multi-valued attributes can be nested arbitrarily.
  - **For example** : the BANGCAP attribute of HOCVIEN is a composite attribute denoted by

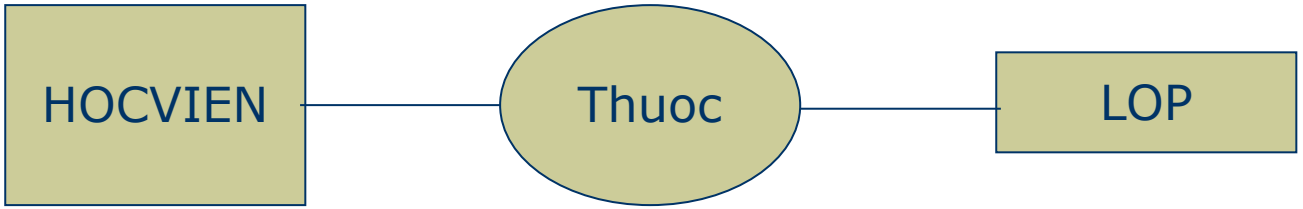
{BANGCAP(TRUONGCAP, NAM, KETQUA, CHUYENNGANH)}

3.4. Key of entity type

- The key of an entity type is an attribute that **identifies** the entity.
- Based on the value of the key, an entity can be **uniquely identified.**
- **For example** :
  - Each student has a unique ID code => The key of the **HOCVIEN entity type** is **Student ID**

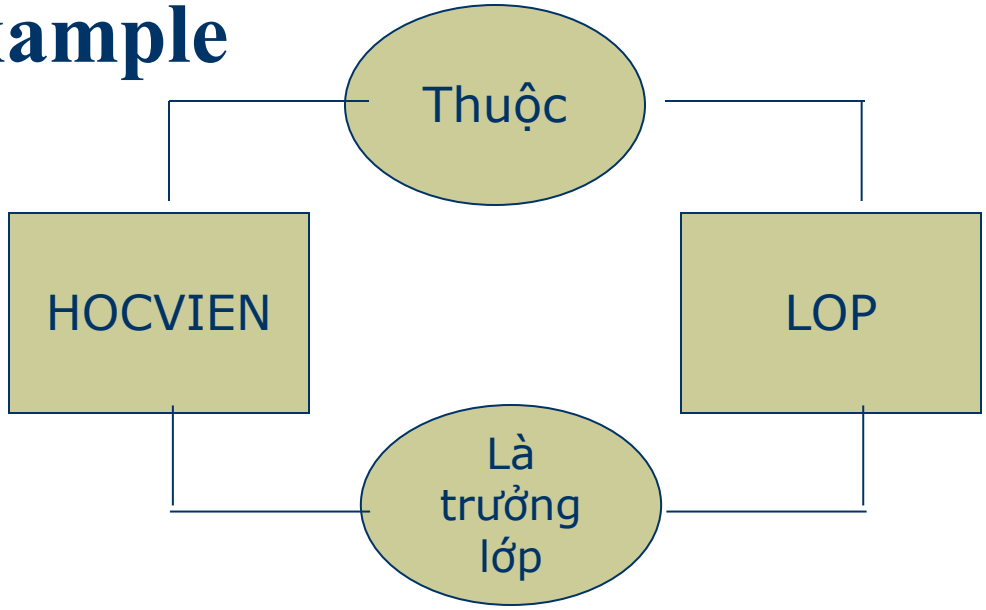
3.5 Relationship type (1)

- **Definition** : a relationship type is a link between two or more entity types.
- **For example** : between two entity types HOCVIEN and LOP there is a relationship type Thuoc
- **Symbol** : an oval or diamond



3.5 Relationship type (2)

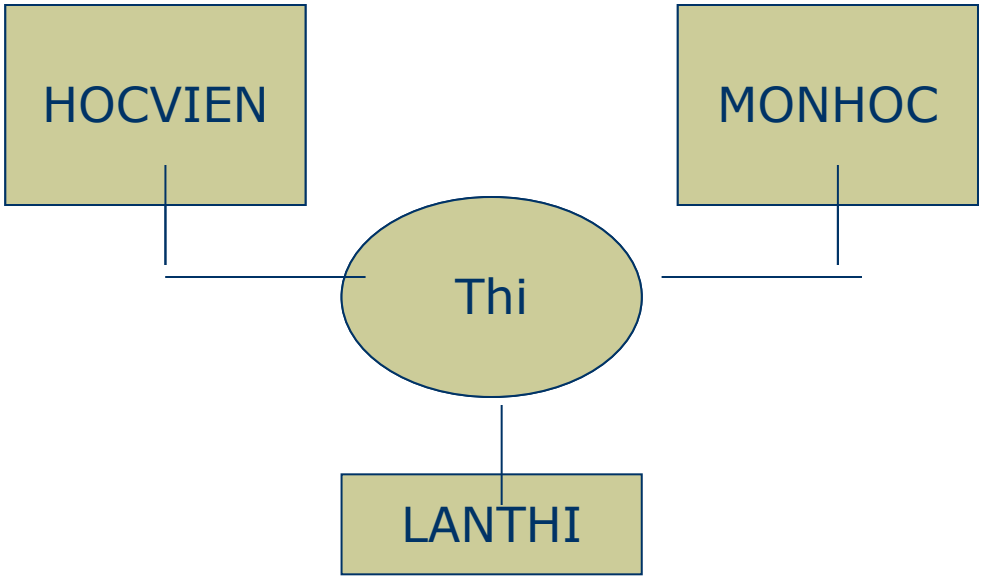
- More than one relationship type can exist between two entity types.
- **For example**



3.5 Relationship degree

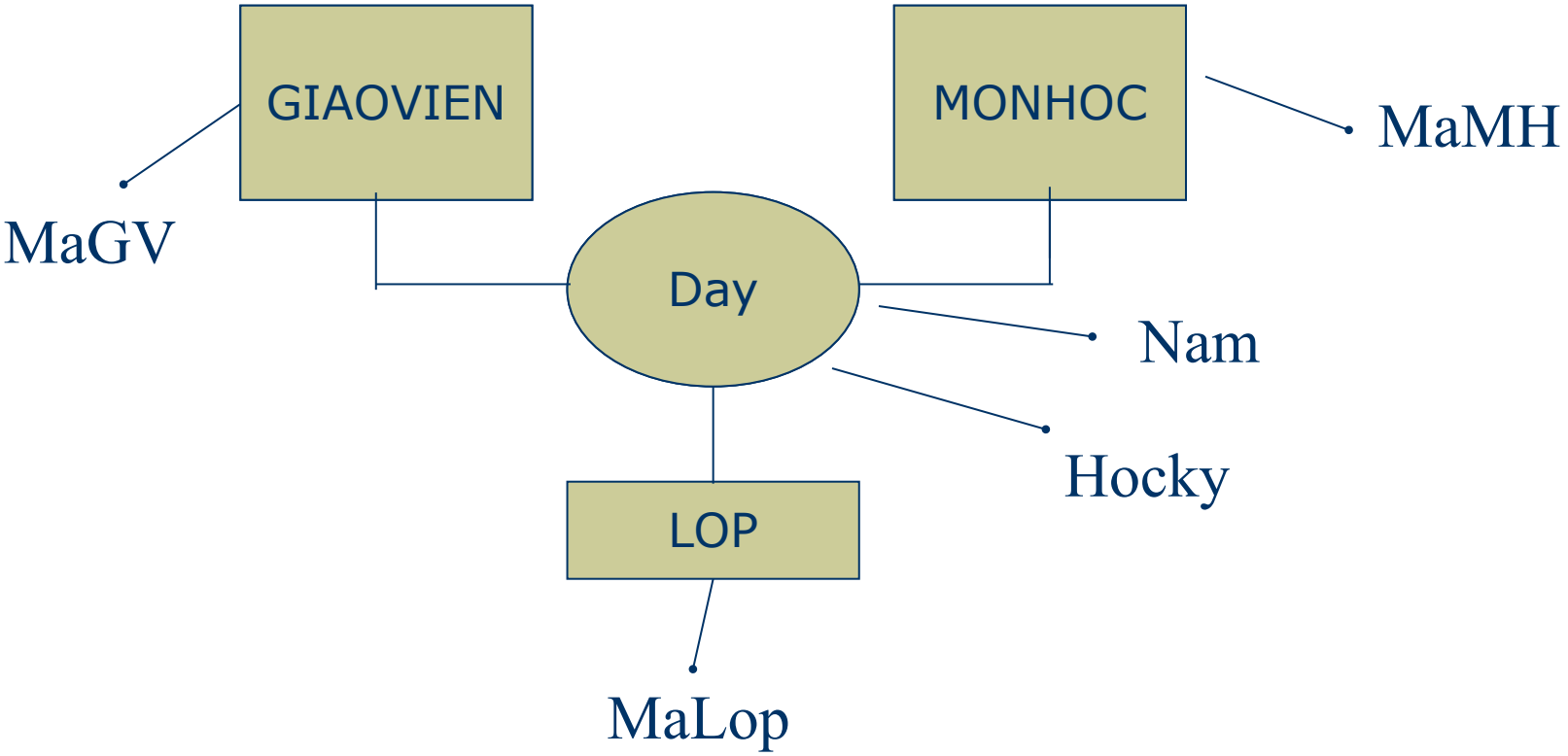
- Relationship degree is the number of entity types participating in that relationship type.
- **Example 1** : Relationship type **Thuoc** has degree 2 because it connects 2 entity types HOCVIEN and LOP.
- **Example 2** : Relationship type **Thi** has degree 3 because it connects 3 entity types LANTHI, HOCVIEN, MONHOC.

3.5 Relationship degree



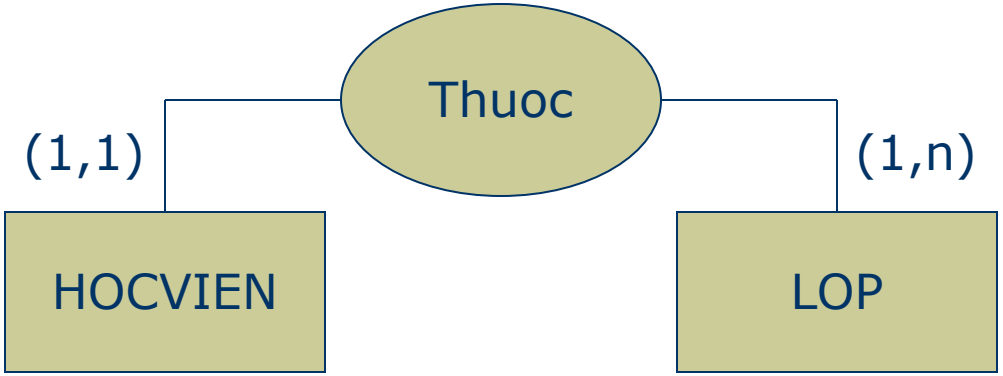
3.6 Attributes of relationship type

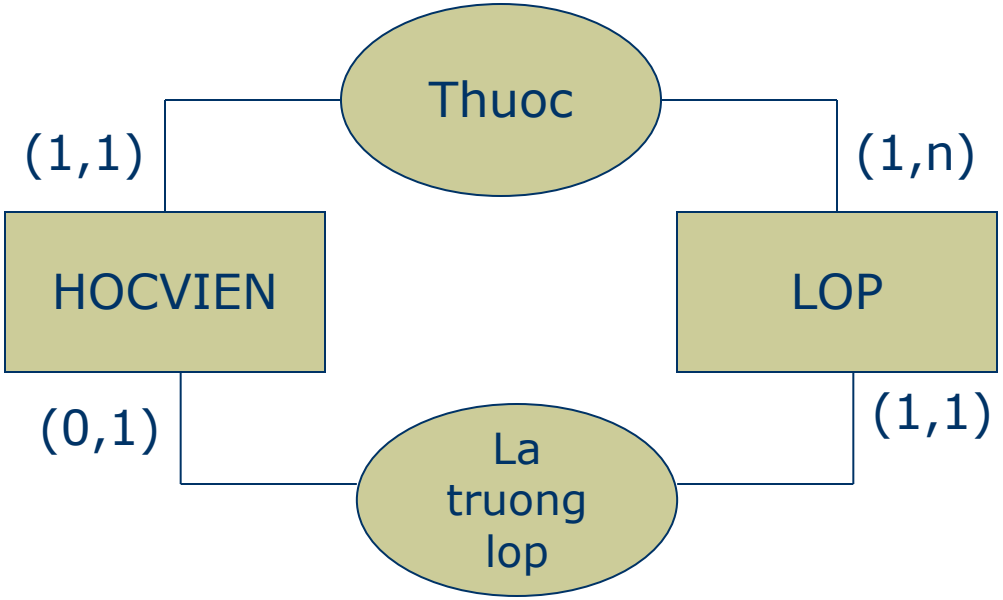
- The attributes of a relationship type include the **key attributes of the entity types participating** in that relationship type. There may also be additional attributes.
- **For example** : The relationship type **Thi** between three entity types HOCVIEN, MONHOC and LANTHI has the attributes Mahv, Mamh, Lanthi, in addition to the separate attributes Diem, Ngaythi


GIAOVIEN
MONHOC
MaMH
MaGV
Day
Nam
Hocky
LOP
MaLop

3.7 Relationship cardinality

- The relationship type represents the links between entities, each link is called a branch.
- **Definition** : the **cardinality** of a branch is the minimum and maximum number of entities belonging to that branch that participate in the relationship type.
- **Symbol** : (minimum quantity, maximum quantity)


Thuoc
(1,1)
(1,n)
HOCVIEN
LOP


Thuoc
(1,1)
(1,n)
HOCVIEN
LOP
(0,1)
(1,1)
La truong lop

3.7 Extended ER model

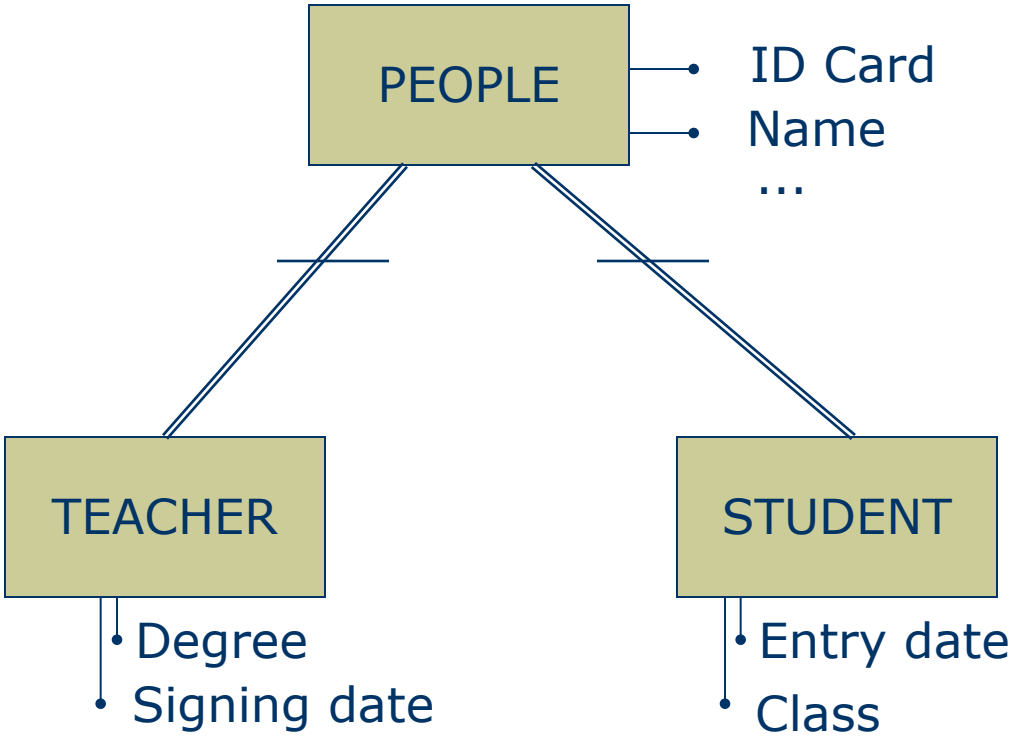
3.7.1 Specialization / Generalization

3.7.2 Recursive relationship

3.7.3 Weak entity type

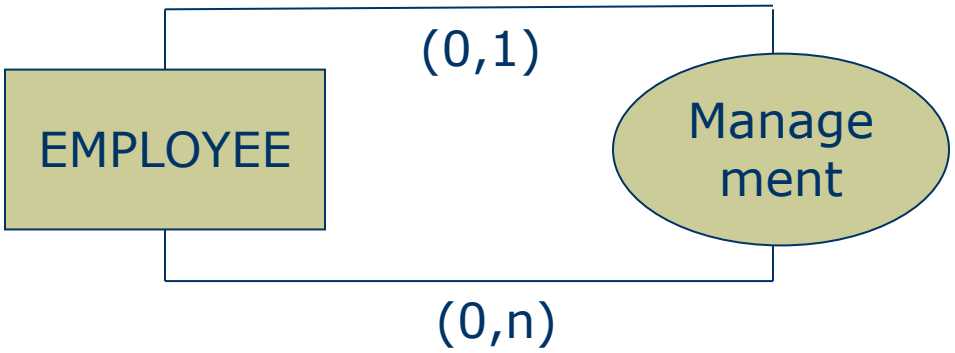
3.7.4 Extended relationship

3.7.1 Specialization (generalization)



3.7.2 Recursive relationship

- **Definition** : is a type of relationship formed from the same type of entity (i.e., an entity type that has a relationship type with itself)
- **For example** : Each employee has a direct manager and that manager is also an employee.

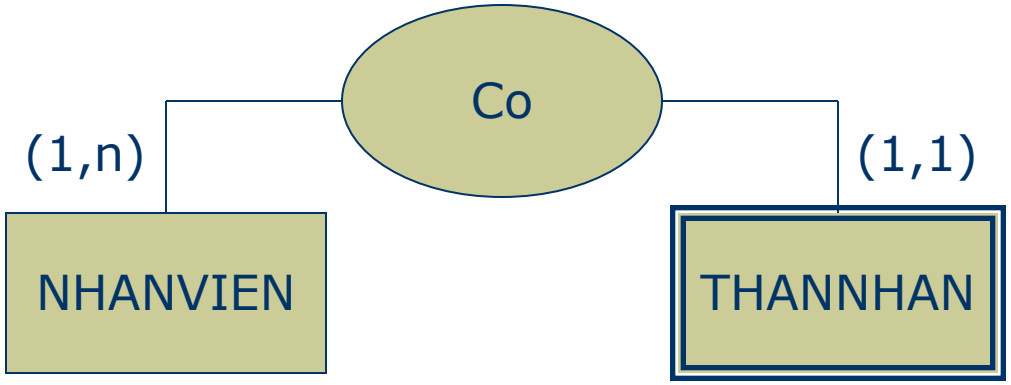


3.7.3 Weak entity type

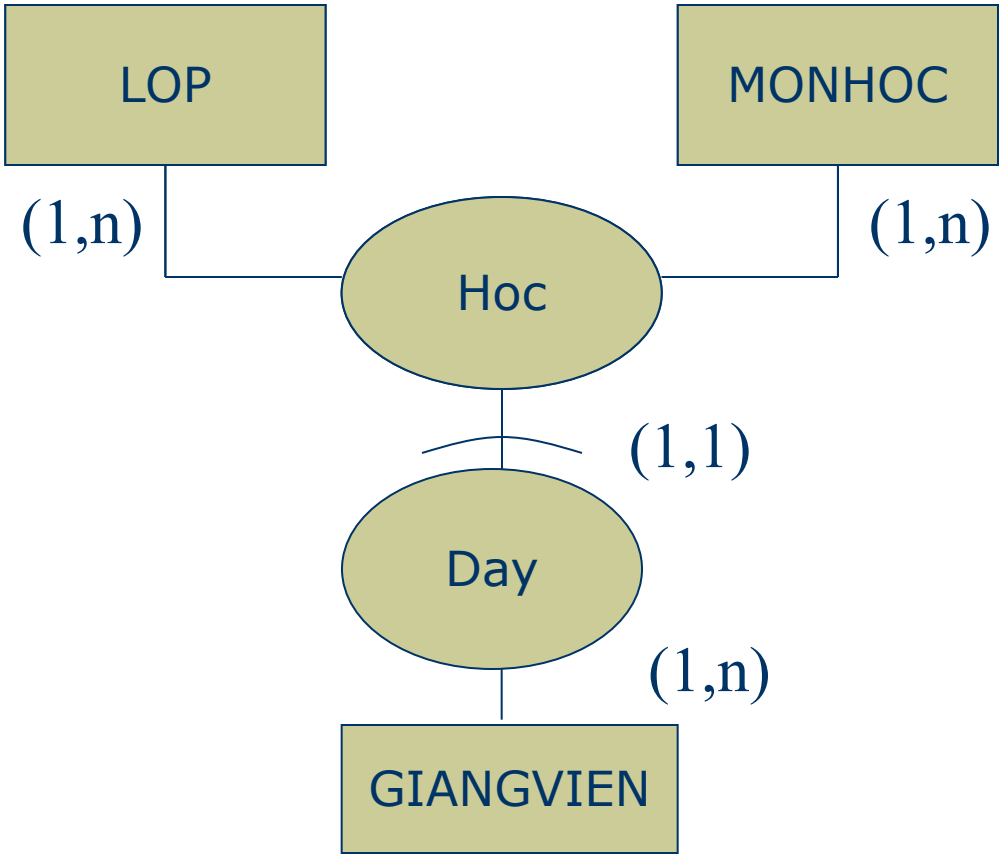
- **Define:**
  - Is an entity type that has **no key attributes**
  - Must participate in an **identifying relationship type** in which there is a host entity type.
- **Symbol** :

- **For example** : THANNHAN entity type is a weak entity type with attributes *Stt, Hoten, Ngsinh, Quanhe* and participates in the relationship type **Co** with NHANVIEN entity type.

3.7.3 Weak entity type



3.7.4 Extended relationship



ER Model Building Exercise

- Building an ER model for the academic management database with the following functions:
  - Information storage: Students, Lecturers, Subjects
  - Place students into class, choose a class monitor
  - Teaching assignment: which class and subject the lecturer teaches, in which semester and school year.
  - Store exam results: which subject the student took the exam for, which trial time, and what score.